

Total No. of Questions : 8]

SEAT No. :

P3110

[5154]-677

[Total No. of Pages : 3

B.E.(Computer Engineering)

DATA MINING TECHNIQUES AND APPLICATIONS
(2012 Pattern) (Semester-I) (410444D) (End Sem.) (Elective-I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1) or Q2), Q3) or Q4), Q5) or Q6), Q7) or Q8).
- 2) Neat diagrams should be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What are the different data normalization methods? Explain them in brief. [6]

b) Consider the training examples shown in the table below for a binary classification problem. [6]

Instance	A1	A2	Class
1	T	T	Yes
2	T	T	Yes
3	T	F	No
4	F	F	Yes
5	F	T	No
6	F	T	No
7	F	F	No
8	T	F	Yes
9	F	T	No

- i) What is the entropy of this collection of training examples with respect to the 'Yes' class
- ii) What are the information gains of A1 and A2 relative to these training examples?
- c) Explain with suitable example the frequent item set generation in Apriori algorithm. [8]

OR

P.T.O.

- Q2) a) What is data preprocessing? Explain the different steps in data preprocessing. [6]
- b) Explain with example K-Nearest-Neighbor Classifier. [6]
- c) Explain the following terms: [8]
- Support count
 - Support
 - Frequent itemset
 - Closed itemset.

- Q3) a) What are interval-scaled variables? Describe the distance measures that are commonly used for computing the dissimilarity of objects described by such variables. [8]
- b) What is meant by complete link hierarchical clustering? [6]
- c) Consider the following vectors x and y. $x=[1,1,1,1]$ $y=[2,2,2,2]$. Calculate: [3]
- Cosine Similarity
 - Euclidean distance.

OR

- Q4) a) Explain with suitable example K-medoids algorithm. [8]
- b) Differentiate between the following: [6]
- Partitioning and hierarchical clustering
 - Centroid and average link hierarchical clustering
 - Symmetric and asymmetric binary variables.
- c) How the Manhattan distance between the two objects is calculated? [3]

- Q5) a) What is Web content mining? Explain in brief. [7]
- b) Assume 'd' is the set of documents and 't' is the term. Write the formulas to determine. [8]
- Term frequency $\text{freq}(d, t)$
 - Weighted term frequency $\text{TF}(d, t)$
 - Inverse document frequency $\text{IDF}(t)$
 - TE-IDF measure $\text{TF-IDF}(d, t)$
- c) What is Web crawler? [2]

OR

- Q6)** a) Compare the different text mining approaches. [9]
b) Explain the following terms: [8]
i) Recommender system
ii) Inverted index
iii) Feature vector
iv) Signature file.

- Q7)** a) Explain with neat diagram systematic machine learning framework. [8]
b) Write short notes on: [8]
i) Big data
ii) Multi-perspective decision making.

OR

- Q8)** a) What is reinforcement learning? Explain. [8]
b) Write short notes on: [8]
i) Wholistic learning
ii) Machine learning

